

Training Course on Multi-criteria decision making in sustainable finance

(4EC, RPTU Kaiserslauten-Landau, Fraunhofer ITWM Kaiserslautern)

Online event

Time: October 26-28, 9:00-16:30

Lecturers: MSc Yannick Becker (YB), Dr. Pascal Halffmann (PH), Prof. Dr. Ralf Korn (RK)

Schedule and Contents:

	Monday (RK)	Tuesday (PH, RK)	Wednesday (RK, YB)
09:00	Introduction & Intro. Continuous-Time Portfolio Optimization (RK)	Introduction to Multiobjective Optimization (PH)	Portfolio Optimization under Uncertainty (YB)
10:15	Coffee break	Coffee break	Coffee break
10:30	Optimal Portfolios: Continuous-time approaches (RK)	Multiobjective Optimization Algorithms and Decision Support (PH)	Optimal Portfolios - Worst Case Approach (RK)
12:00	Lunch	Lunch	Lunch
13:30	Optimal Portfolios: Optimal Portfolios with Sustainability Constraints and Climate Scenarios (RK)	Applications: CPPI-Strategies(RK) Non-hedgeable risks (RK) Practical aspects (RK)	Solvency II and Optimal Portfolios (RK)
15:00	Coffee break	Coffee break	Coffee break
15:30	Exercise Session (RK)	Exercise Session (TP)	Final discussion, summary and closing
16:30			

About the training course:

The three-day course focuses on applying **dynamic and static optimization techniques** in discrete- and continuous-time **market models**. As special applications we will consider the inclusion of sustainability aspects and multicriteria settings in our framework.

Students will see and explore (in special exercise classes) the applications of the optimization technologies in

- Modelling and solution of optimal portfolio selection problems in the presence of climate scenarios
- Inclusion of digital assets into standard portfolio problems in different ways
- Further applications for sensitivity analyses

The focus of the course will be on both algorithms and modelling. The students will be equipped with hands on (but rigorous) strategies to set up and solve dynamic and multicriteria portfolio problems. They will be encouraged to apply advanced techniques such as worst-case control techniques to problems in digital asset markets and to learn the corresponding fundamentals of continuous-time finance by focusing on the application. Doing (partly supervised) exercises will also be an essential part of the course.

Prior knowledge:

For a successful participation in the course some prior knowledge in financial mathematics and in Itô-calculus would be desirable. The necessary facts from stochastic calculus are provided in advance in a downloadable file „MSCA_Prelimiaries.pdf“ . Further necessary details or requests can be dealt with in the exercise sections during the workshop.